mycode

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0.1 Information Retrieval Component: Individual Project

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###Introduction In this project we are required to index and query a WES data set. The project is divided into 2 sepearte parts the Indexing and the Querying.

###Part 1: Indexing I created a python code box for importing all the necessary libraries. To load and read the data in XMLs raw tags I decided to use REGEX. This is the abbrivation of regualr expression.l I chose this as it is a process I am familiar with. I ouputted the number of documnets extracted to maintain engineering hygeine: Extracted 331 documents

I then tokenized the document and output a sample to ensure the code was logically correct: Document 1 tokens: [‘<’, ‘!’, ‘[’, ‘CDATA’, ‘[’, ’Joseph’, ’Nicollet’, ’and’, ’the’, ’Upper’, ’Mississippi’, ’River’, ’.’, ’On’, ’July’, ’24’, ’,’, ’1786’, ’,’, ’French’] Document 2 tokens: [’<‘,’¡,’[‘, ’CDATA’, ’[‘, ’Beaumarchais’, ‘and’, ‘Figaro’, “‘s”, ’Wedding’, ‘:’, ‘On’, ‘May’, ‘18’, ‘,’, ‘1799’, ‘,’, ‘French’, ‘playwright’, ‘Pierre-Augustin’] Preprocessing done Original document: <![CDATA[Joseph Nicollet and the Upper Mississippi River.

Lastly I built the terms by document matrix containing the TF.IDF weight for eahc term within the document On July 24, 1786, French geographer, astr … Preprocessed tokens: [‘cdata’, ‘joseph’, ‘nicollet’, ‘upper’, ‘mississippi’, ‘river’, ‘juli’, ‘24’, ‘1786’, ‘french’, ‘geograph’, ‘astronom’, ‘mathematician’, ‘joseph’, ‘nicola’, ‘nicollet’, ‘born’, ‘best’, ‘known’, ‘map’] … aachen abadon abandon abbasid abbevil abbey abbot abbé abc
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And that is Part 1 completed.

###Part 2: Querying I set the query to “French geographer goes missing in the alps” I then preprocessed the query by passing it through the tokenization, case-folding, stop-word removal and stemming models created in the indexing part of the code. This was the succesful output:

French geographer goes missing in Alps Original query: French geographer goes missing in Alps Preprocessed query tokens: [‘french’, ‘geograph’, ‘goe’, ‘miss’, ‘alp’]

I then used cosine similarity to calculate the similarity between the query i created and each document. This was the result: Top 5 relevant documents Document 204 with similarity score 0.0792 <![CDATA[Louis Antoine de Bougainville and his Voyage Around the World.

Bougainville reaching Tahiti Probably on November 11, 1729, French admiral and explorer Louis Antoine de Bougainville was born. …

Document 326 with similarity score 0.0664 <![CDATA[Claude Joseph Rouget de Lisle and the Marseillaise.

Claude Joseph Rouget de Lisle singing La Marseillaise. On April 25, 1792, French army officier Claude Joseph Rouget de Lisle during the F …

Document 192 with similarity score 0.0571 <![CDATA[John Tyndall and the Physics of Air.

John Tyndall (1820-1893). caricatured as a preacher in the magazine Vanity Fair, 1872. On August 2, 1820, British physicist John Tyndall was born. His i …

Document 250 with similarity score 0.0532 <![CDATA[Abraham Ortelius and the Theatrum Orbis Terrarum. …

Erich von Drygalski (1865-1949). . On February 9, 1865, German geographer, geophysicist and polar scientist Erich Dagobert von Drygalski was bo …

I then outputted the top 5 relevant documents: Ranked list of documents based on query similarity:

1. Document 204 (similarity score: 0.0792) <![CDATA[Louis Antoine de Bougainville and his Voyage Around the World.

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2. Document 326 (similarity score: 0.0664) <![CDATA[Claude Joseph Rouget de Lisle and the Marseillaise.

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4. Document 250 (similarity score: 0.0532) <![CDATA[Abraham Ortelius and the Theatrum Orbis Terrarum.

Ortelius World Map – Typus Orbis Terrarum 1570. On May 20, 1570, Belgian cartographer and geographer Abraham Ortelius publishes the first m …

5. Document 113 (similarity score: 0.0528) <![CDATA[Erich von Drygalski’s Antarctic Expeditions.

Erich von Drygalski (1865-1949). . On February 9, 1865, German geographer, geophysicist and polar scientist Erich Dagobert von Drygalski was bo …

###Conclusion
Overall this assignment was simple and strait to the point. The results from my code were as expected and there are minimal logical errors.

[49]:
```
#importing
import nltk
import os
import re
import sys
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

print("Imports done")
```

```
Imports done
```

[ ]:
```
#PART 1: INDEXING
#first load the data and read the docoment
folder_path = 'docs'

all_docs = []
for filename in os.listdir(folder_path):
    file_path = os.path.join(folder_path, filename)


    if os.path.isfile(file_path):
        with open(file_path, 'r', encoding='utf-8') as f:
            data = f.read()


            matches = [m.strip() for m in re.findall(r'<raw>(.*?)</raw>', data,␣
 ↪re.DOTALL | re.IGNORECASE)]
            all_docs.extend(matches)

print(f"Extracted {len(all_docs)} documents")

#tokenisation
nltk.download('punkt')

tokenize_docs= [word_tokenize(doc) for doc in all_docs]

#output first 2 tokenised documents
for i, tokens in enumerate(tokenize_docs[:2], 1):
    print(f"Document {i} tokens: {tokens[:20]}")  # show first 20 tokens
```

```
#extraction of index terms
nltk.download('stopwords', quiet=True)
nltk.download('punkt', quiet=True)

stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()

preprocessed_docs = []

for doc in all_docs:
    doc_lower=doc.lower()
    tokens = nltk.word_tokenize(doc_lower)
    tokens_nostop = [word for word in tokens if word.isalnum() and word not in␣
 ↪stop_words]
    stemmed_tokens = [stemmer.stem(word) for word in tokens_nostop]
    preprocessed_docs.append(stemmed_tokens)
print("Preprocessing done")

#example
print("Original document:", all_docs[0][:100], "...")
print("Preprocessed tokens:", preprocessed_docs[0][:20], "...")

#weighting of index terms
filtered_docs = [[token for token in doc if token.isalpha()] for doc in␣
 ↪preprocessed_docs]
docs_as_strings = [' '.join(doc) for doc in preprocessed_docs]
vectorizer = TfidfVectorizer(
    tokenizer=lambda x: x,
    preprocessor=lambda x: x,
    token_pattern=None
)
tfidf_matrix = vectorizer.fit_transform(filtered_docs)
terms = vectorizer.get_feature_names_out()
tdm_df = pd.DataFrame(tfidf_matrix.toarray(), columns=terms)
print(tdm_df.iloc[:5, :10])
```

```
Extracted 331 documents

[nltk_data] Downloading package punkt to /root/nltk_data…
[nltk_data]   Package punkt is already up-to-date!

Document 1 tokens: ['<', '!', '[', 'CDATA', '[', 'Joseph', 'Nicollet', 'and',
'the', 'Upper', 'Mississippi', 'River', '.', 'On', 'July', '24', ',', '1786',
',', 'French']
Document 2 tokens: ['<', '!', '[', 'CDATA', '[', 'Beaumarchais', 'and',
'Figaro', "'s", 'Wedding', '.', 'On', 'May', '18', ',', '1799', ',', 'French',
'playwright', 'Pierre-Augustin']
```

```
Preprocessing done
Original document: <![CDATA[Joseph Nicollet and the Upper Mississippi River.

On July 24, 1786, French geographer, astr …
Preprocessed tokens: ['cdata', 'joseph', 'nicollet', 'upper', 'mississippi',
'river', 'juli', '24', '1786', 'french', 'geograph', 'astronom',
'mathematician', 'joseph', 'nicola', 'nicollet', 'born', 'best', 'known', 'map']
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0     0.0     0.0  0.000000      0.0      0.0    0.0    0.0   0.0  0.0
1     0.0     0.0  0.024862      0.0      0.0    0.0    0.0   0.0  0.0
2     0.0     0.0  0.000000      0.0      0.0    0.0    0.0   0.0  0.0
3     0.0     0.0  0.000000      0.0      0.0    0.0    0.0   0.0  0.0
4     0.0     0.0  0.000000      0.0      0.0    0.0    0.0   0.0  0.0

   abdomin
0      0.0
1      0.0
2      0.0
3      0.0
4      0.0
```

[59]:
```python
#Part 2: Querying
#create query
query= "French geographer goes missing in Alps"
print (query)

#preprocess query
query_lower = query.lower()
query_tokens = nltk.word_tokenize(query_lower)
query_nostop = [word for word in query_tokens if word.isalnum() and word not in␣
 ↪stop_words]
query_stemmed = [stemmer.stem(word) for word in query_nostop]
print("Original query:", query)
print("Preprocessed query tokens:", query_stemmed)

#csine similarity between query and docs
query_vec = vectorizer.transform([query_stemmed])
similarity=cosine_similarity(query_vec, tfidf_matrix).flatten()
ranked_indices = similarity.argsort()[::-1]
print("Top 5 relevant documents:")
for idx in ranked_indices[:5]:
    print(f"Document {idx} with similarity score {similarity[idx]:.4f}")
    print(all_docs[idx][:200], "...\n")
    print()

#printing top 5 documents on cosine similiarity
```

```
print("Ranked list of documents based on query similarity:\n")
for rank, idx in enumerate(ranked_indices, 1):
    print(f"{rank}. Document {idx} (similarity score: {similarity[idx]:.4f})")
```

```
French geographer goes missing in Alps
Original query: French geographer goes missing in Alps
Preprocessed query tokens: ['french', 'geograph', 'goe', 'miss', 'alp']
Top 5 relevant documents:
Document 204 with similarity score 0.0792
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4. Document 250 (similarity score: 0.0532)
5. Document 113 (similarity score: 0.0528)
6. Document 139 (similarity score: 0.0521)
7. Document 196 (similarity score: 0.0492)
8. Document 1 (similarity score: 0.0435)
9. Document 287 (similarity score: 0.0406)
10. Document 0 (similarity score: 0.0360)
11. Document 289 (similarity score: 0.0351)
12. Document 322 (similarity score: 0.0291)
13. Document 143 (similarity score: 0.0287)
14. Document 179 (similarity score: 0.0287)
15. Document 74 (similarity score: 0.0258)
16. Document 82 (similarity score: 0.0253)
17. Document 272 (similarity score: 0.0224)
18. Document 286 (similarity score: 0.0218)
19. Document 320 (similarity score: 0.0218)
20. Document 152 (similarity score: 0.0217)
21. Document 59 (similarity score: 0.0215)
22. Document 48 (similarity score: 0.0211)
23. Document 183 (similarity score: 0.0210)
24. Document 153 (similarity score: 0.0207)
25. Document 189 (similarity score: 0.0194)
26. Document 95 (similarity score: 0.0188)
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38. Document 232 (similarity score: 0.0151)
39. Document 67 (similarity score: 0.0149)
40. Document 6 (similarity score: 0.0132)
41. Document 239 (similarity score: 0.0131)
42. Document 249 (similarity score: 0.0122)
43. Document 235 (similarity score: 0.0116)
44. Document 321 (similarity score: 0.0115)
45. Document 35 (similarity score: 0.0113)
46. Document 271 (similarity score: 0.0113)
47. Document 43 (similarity score: 0.0111)
48. Document 131 (similarity score: 0.0108)
49. Document 37 (similarity score: 0.0104)
50. Document 157 (similarity score: 0.0101)
51. Document 247 (similarity score: 0.0097)

52. Document 16 (similarity score: 0.0093)
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280. Document 60 (similarity score: 0.0000)
281. Document 58 (similarity score: 0.0000)
282. Document 93 (similarity score: 0.0000)
283. Document 65 (similarity score: 0.0000)
284. Document 54 (similarity score: 0.0000)
285. Document 55 (similarity score: 0.0000)
286. Document 49 (similarity score: 0.0000)
287. Document 52 (similarity score: 0.0000)
288. Document 78 (similarity score: 0.0000)
289. Document 77 (similarity score: 0.0000)
290. Document 75 (similarity score: 0.0000)
291. Document 79 (similarity score: 0.0000)

292. Document 28 (similarity score: 0.0000)
293. Document 27 (similarity score: 0.0000)
294. Document 26 (similarity score: 0.0000)
295. Document 25 (similarity score: 0.0000)
296. Document 22 (similarity score: 0.0000)
297. Document 21 (similarity score: 0.0000)
298. Document 36 (similarity score: 0.0000)
299. Document 92 (similarity score: 0.0000)
300. Document 34 (similarity score: 0.0000)
301. Document 32 (similarity score: 0.0000)
302. Document 62 (similarity score: 0.0000)
303. Document 63 (similarity score: 0.0000)
304. Document 284 (similarity score: 0.0000)
305. Document 283 (similarity score: 0.0000)
306. Document 282 (similarity score: 0.0000)
307. Document 319 (similarity score: 0.0000)
308. Document 279 (similarity score: 0.0000)
309. Document 278 (similarity score: 0.0000)
310. Document 277 (similarity score: 0.0000)
311. Document 308 (similarity score: 0.0000)
312. Document 307 (similarity score: 0.0000)
313. Document 305 (similarity score: 0.0000)
314. Document 304 (similarity score: 0.0000)
315. Document 281 (similarity score: 0.0000)
316. Document 15 (similarity score: 0.0000)
317. Document 14 (similarity score: 0.0000)
318. Document 61 (similarity score: 0.0000)
319. Document 13 (similarity score: 0.0000)
320. Document 44 (similarity score: 0.0000)
321. Document 42 (similarity score: 0.0000)
322. Document 41 (similarity score: 0.0000)
323. Document 328 (similarity score: 0.0000)
324. Document 39 (similarity score: 0.0000)
325. Document 38 (similarity score: 0.0000)
326. Document 291 (similarity score: 0.0000)
327. Document 292 (similarity score: 0.0000)
328. Document 290 (similarity score: 0.0000)
329. Document 31 (similarity score: 0.0000)
330. Document 30 (similarity score: 0.0000)
331. Document 46 (similarity score: 0.0000)